QuickBite Platform

Formal UML 2.5 Use Case Diagram Specification & Functional Blueprint

Standard: OMG UML 2.5 Specification
Scope: System & Business High-Level
Target Arch: Electron 30 + Spring Boot 3 + Supabase
Version: 1.0 (Production Verified)

PRIMARY HUMAN ACTORS	SUPPORTING SYSTEMS	TOTAL USE CASES	SUBSYSTEM PACKAGES
Buyer, Shop Owner, Courier, Admin	**Firebase Auth, Razorpay, Supabase**	**28 Comprehensive Capabilities**	**7 Functional Boundaries**

Executive Architectural Intent: This specification provides the complete UML 2.5 Use Case architecture of the QuickBite food marketplace and delivery system. Modeled at the system and business functional level, it defines the complete functional boundary between external actors and platform capabilities through formal UML associations, specialization/generalization, «include» subroutines, and conditional «extend» extension points without getting bogged down in low-level code implementation details.

1. Formal UML 2.5 Use Case Notation & Legend

The table below summarizes the formal OMG UML 2.5 visual notations utilized across all system diagrams:

UML ELEMENT	VISUAL NOTATION	UML 2.5 SEMANTICS	QUICKBITE PLATFORM APPLICATION
Actor (Human)	Stick figure with label	Represents a specific external user role that initiates or participates in a use case.	`Buyer`, `Shop Owner`, `Delivery Partner`, `Platform Admin`, `Guest`.
Actor (System)	Box with «system»	An external secondary software service that supplies authentication, payment, or storage.	`Firebase Auth Service`, `Razorpay Gateway`, `Supabase Cloud Storage`, `Google Maps`.
Use Case	Horizontal Ellipse	A discrete unit of functional system behavior yielding an observable value to an actor.	Named with active verb-noun phrases (e.g., *Browse Approved Shops*, *Place Order*, *Claim Delivery*).
System Boundary	Labeled Box `QuickBite`	Delineates internal platform capabilities from external actors and foreign services.	Encloses all 7 internal subsystems; external services and actors remain strictly outside.
Association	Solid Line (——)	Communication path connecting an actor directly to the use cases they participate in.	Links Buyer to *Browse Items*, Courier to *Claim Delivery*, Admin to *Review Shop*.
«include»	Dashed Arrow `- - ->`	Mandatory dependency: the base use case explicitly includes the behavior of the target.	*Place Order* «`include`» *Process Payment* & *Reserve Stock Atomically*.
«extend»	Dashed Arrow `<- - -`	Conditional extension: target use case augments base behavior under defined conditions.	*Cancel Order* «`extend`» *Track Order Status* (Condition: prior to kitchen prep).
Generalization	Solid Line + Hollow Arrow	Specialization inheritance: derived actor or use case inherits properties of parent.	`Buyer`, `Shop Owner`, `Courier`, `Admin` specialize `Registered User`.

2. Architectural Boundary & Packaging Overview

The system functional surface is structured into 7 cohesive subsystem packages:

1. Identity & Access Control: Registration, Firebase verification, and onboarding gate.
2. Marketplace & Discovery: Shop browsing, search, and cold-start recommendation.
3. Cart & Order Placement: Single-shop cart, atomic checkout, and payment.
4. Kitchen Operations: Store profile, menu catalog, and food preparation.
5. Courier Logistics: Claiming queue, map routing, and OTP customer handover.
6 & 7. Governance & Auditing: Approvals, disputes, refunds, and audit trail logs.

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Actor Catalog & Supporting Systems Hierarchy

Document Section: Actors & External Integrations
Classification: Human & System Roles

3. Human Actor Catalog & Inheritance Hierarchy

All authenticated personas in QuickBite specialize the general `Registered User` abstraction, inheriting fundamental authentication, session lifecycle, and profile management capabilities while extending role-specific responsibilities:

ACTOR	HIERARCHY	PRIMARY PURPOSE	KEY USE CASES PARTICIPATED IN
Guest	Unregistered	Unauthenticated campus visitor exploring food options.	*Browse Approved Shops*, *Multi-Criteria Search*, *Sign Up with Role Details*.
Registered User	`«abstract»` Base Actor	Parent of all authenticated user types on the platform.	*Sign In with Firebase*, *Refresh Session Token*, *Request Account Removal*.
Buyer	Specializes `Registered User`	End customer discovering food, ordering, and receiving meals.	*Manage Single-Shop Cart*, *Place Order*, *Track Status*, *Cancel Order*, *Provide Handover OTP*, *Rate Order*.
Shop Owner	Specializes `Registered User`	Food vendor managing storefront, catalog, and kitchen workflow.	*Register Shop*, *Manage Product Menu*, *Toggle Open/Closed*, *Process Kitchen Orders*, *Request Offboarding*.
Delivery Partner	Specializes `Registered User`	Courier fulfilling orders between food kitchens and customer drop pins.	*Browse Available Queue*, *Claim Delivery*, *Confirm Pickup*, *Verify Buyer OTP*, *Report Issue*, *View Earnings*.
Platform Admin	Specializes `Registered User`	System operator overseeing marketplace safety and dispute mediation.	*Review Shop Approvals*, *Review Courier Approvals*, *Manage Admin Keys*, *Adjudicate Disputes*, *Authorize Refunds*, *Inspect Audit Logs*.

4. Secondary Supporting System Actors

QuickBite integrates with four enterprise external systems that participate as secondary supporting actors:

Firebase Authentication Service — SECURITY

Role: Identity Provider & Token Authority.
Interactions: Issues signed JWT ID tokens on sign-in, validates credential hashes, and provides custom claims synchronization for role-based authorization.

Razorpay Payment Gateway — FINTECH

Role: Payment Settlement Gateway.
Interactions: Creates transactional checkout orders, provides test/live checkout modals, verifies cryptographic HMAC payment signatures, and processes refunds.

Supabase Cloud (PostgreSQL & Storage) — DATA & MEDIA

Role: Relational Persistence & Media CDN.
Interactions: ACID transaction management, row-level pessimistic locking for delivery claiming, and high-resolution photo hosting for shops and menu items.

Google Maps API & Geocoding — GEO-SPATIAL

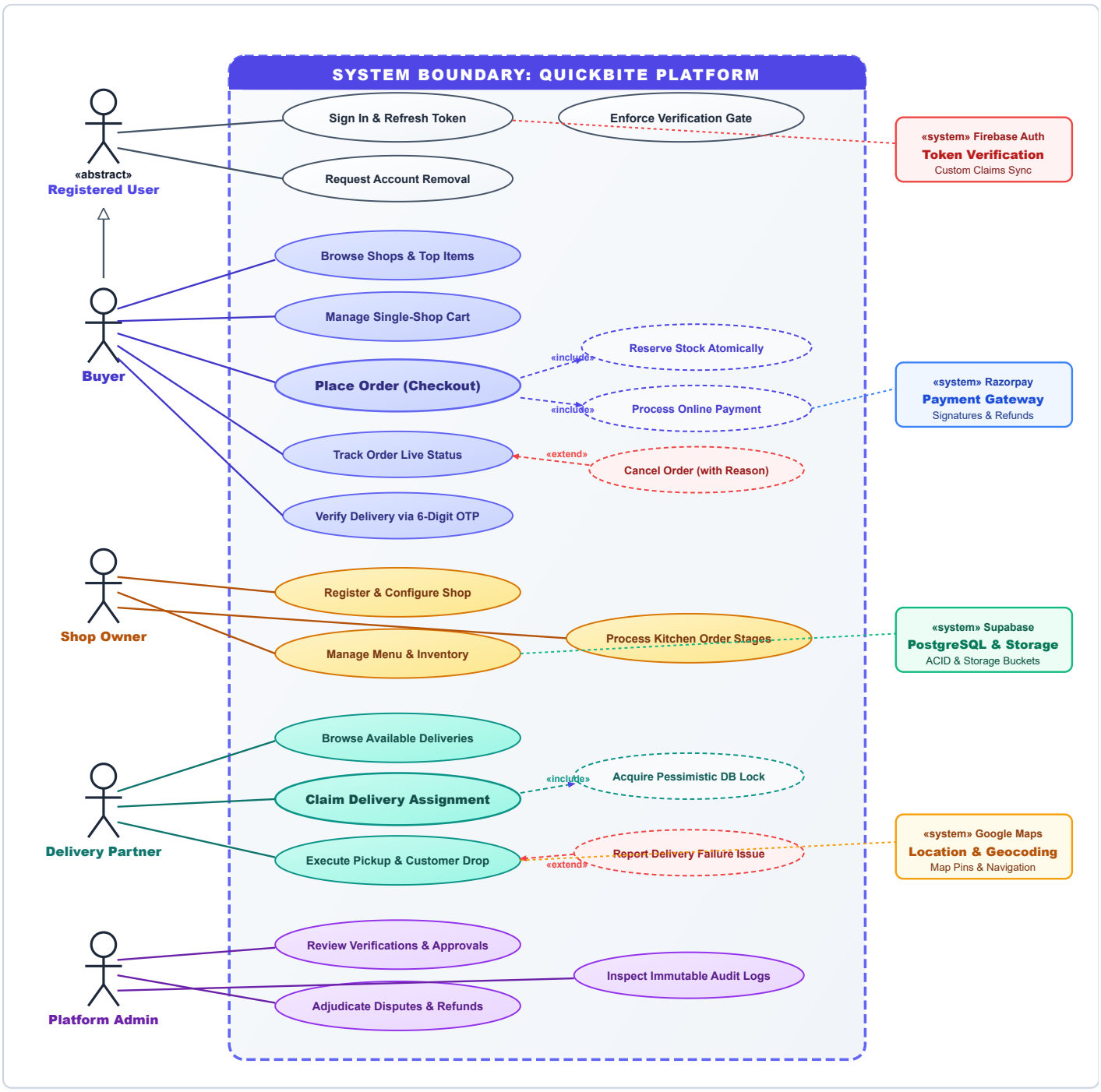
Role: Location Services.
Interactions: Reverse geocoding for delivery address resolution, precise latitude/longitude pinning, and deep links for courier turn-by-turn navigation.

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Master System-Level UML 2.5 Use Case Diagram

Diagram Type: Master Architecture Overview
Orientation: End-to-End System Boundary

The diagram below synthesizes the complete platform boundary, connecting human actors, primary use cases, «include» and «extend» dependencies, and external systems:



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Subsystem Functional Decomposition (Subsystems 1 to 3)

Focus: User Onboarding, Catalog & Commerce
Traceability: Architecture Parity

5. Subsystems 1, 2 & 3: User Onboarding, Marketplace & Ordering

Subsystem 1: Identity, Onboarding & Verification Gate

SECURITY & AUTH

Governs user account creation, client-side Firebase Auth JS SDK token generation, Spring Security `FirebaseTokenFilter` Bearer token verification, custom claims synchronization, and onboarding gating.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-AUTH-01: Sign Up with Role Details	Guest	Firebase Auth	Creates Firebase Auth credentials and posts role profile to `/api/v1/auth/signup`. If backend rejects, rolls back Firebase user immediately.
UC-AUTH-02: Sign In & Refresh Token	Registered User	Firebase Auth	Authenticates credentials with Firebase; frontend sends ID Token to `/api/v1/auth/login`. Backend verifies token, resolves database profile, and synchronizes custom claims.
UC-AUTH-03: Enforce Verification Gate	RolePolicy Guard	Supabase DB	Restricts unapproved `SHOP_OWNER`, `DELIVERY_PARTNER`, and `ADMIN` strictly to `/verification` until Admin approval. Verified accounts open full role dashboards.
UC-AUTH-04: Request Account Removal	Registered User	Platform Admin	Submits formal offboarding request. Does not immediately delete; requires active Admin authorization to ensure all pending orders and settlements are closed.

Subsystem 2: Marketplace Discovery & Recommendations

CATALOG ENGINE

Enables marketplace discovery across all approved storefronts, powering cold-start food feeds and multi-criteria substring search filters.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-MKT-01: Browse Approved Shops	Buyer / Guest	Supabase DB	Loads all active approved shops upfront with open/closed indicators and embedded Google Maps pins.
UC-MKT-02: Search Food & Shops	Buyer / Guest	Supabase DB	Executes combined substring search across product names, shop names, and neighborhood campus locations.
UC-MKT-03: View Personalized Feed	Buyer	Recommendation Engine	Ranks dishes by user taste profile & ratings; applies cold-start algorithm for fresh buyers without history.

Subsystem 3: Cart Management, Checkout & Order Tracking

COMMERCE CORE

Guarantees the single-shop cart invariant, atomic inventory reservation, online payments, delivery OTP generation, and cancellation policies.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-ORD-01: Manage Single-Shop Cart	Buyer	Cart Domain Model	Enforces that all items originate from exactly one vendor. Auto-calculates subtotal and ₹300 free delivery rule.
UC-ORD-02: Place Order (Checkout)	Buyer	Razorpay & Supabase	«include» *Reserve Stock Atomically* (pessimistic DB lock on product stock). «include» *Process Online Payment* (Razorpay test/live checkout modal). Generates unassigned `DeliveryAssignment` and unique 6-digit delivery OTP.
UC-ORD-03: Track Live Order Status	Buyer	Supabase Polling	Real-time polling (every 6s) showing state progression (PLACED → ACCEPTED → PREPARING → READY → OUT_FOR_DELIVERY → DELIVERED).
UC-ORD-04: Cancel Order	Buyer	Dialog Modal	«extend» *Track Live Order Status*. Condition: Order status in `PLACED` or `ACCEPTED`. Restores product inventory and flags refund.
UC-ORD-05: Rate & Review Order	Buyer	Feedback Service	Post-delivery evaluation (rating 1-5 and comment). Constrained by unique constraint (max 1 review per delivered order).

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Subsystem Functional Decomposition (Subsystems 4 & 5)

Focus: Kitchen Fulfillment & Courier Logistics
Concurrency: Pessimistic DB Locking

6. Subsystems 4 & 5: Kitchen Operations & Courier Logistics

Subsystem 4: Shop Management & Kitchen Fulfillment

VENDOR OPERATIONS

Empowers food vendors to maintain their menu catalog, control storefront availability, manage live kitchen orders, and advance meal preparation stages.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-SHOP-01: Register & Configure Shop	Shop Owner	Supabase Storage	Submits shop profile, Leaflet map location coordinates, and storefront photography uploaded to Supabase Storage (`QuickBite/shops`).
UC-SHOP-02: Manage Menu Catalog	Shop Owner	Supabase Storage	Full CRUD for dishes: defines title, price, inventory stock counters, availability toggles, and food pictures (`QuickBite/products`).
UC-SHOP-03: Process Kitchen Orders	Shop Owner	Order Service	Transitions order stages: `PLACED` → `ACCEPTED` → `PREPARING` → `READY`. When marked READY, order becomes claimable by couriers.
UC-SHOP-04: Toggle Storefront Status	Shop Owner	Marketplace Service	Instant Open/Closed switch. Closed storefronts remain browseable in catalog but block new checkout placement.

Subsystem 5: Courier Logistics & Delivery Fulfillment

COURIER LOGISTICS

Coordinates order assignment, concurrency-safe delivery claiming, GPS navigation routing, and OTP-verified customer handovers.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-DEL-01: Browse Available Deliveries	Delivery Partner	Supabase DB	Polls unassigned orders in `ACCEPTED`, `PREPARING`, or `READY` state with pickup and drop location previews.
UC-DEL-02: Claim Delivery Assignment	Delivery Partner	Supabase DB	«`include`» *Acquire Pessimistic DB Lock*. Executes atomic claim query under `READ_COMMITTED` transaction. If contested, surfaces `409 Conflict` cleanly and refreshes board.
UC-DEL-03: Confirm Kitchen Pickup	Delivery Partner	Order Service	Marks assignment `PICKED_UP` and updates order to `OUT_FOR_DELIVERY`, exposing the 6-digit delivery OTP to buyer.
UC-DEL-04: Execute Delivery Handover	Delivery Partner	Buyer	«`include`» *Verify Buyer 6-Digit OTP*. Requires entry of the secret OTP held by the buyer. Successful match transitions order to `DELIVERED` and credits ₹30 trip earnings.
UC-DEL-05: Report Delivery Issue	Delivery Partner	Admin Service	«`extend`» *Execute Delivery Handover*. Triggered when customer is unreachable or address is invalid, recording explanation reason for admin dispute adjudication.
UC-DEL-06: Review History & Earnings	Delivery Partner	Delivery Service	Displays fulfilled deliveries list and calculates aggregate earnings based on standard ₹30 trip fee per delivery.

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Subsystem Functional Decomposition (Subsystems 6 & 7)

Focus: Platform Governance, Disputes & Audit Trail
Compliance: Non-Repudiable Logs

7. Subsystems 6 & 7: Governance, Disputes, Refunds & Audit Trail

Subsystem 6: Platform Governance, Disputes & Refunds

GOVERNANCE CORE

Enforces platform integrity through partner vetting, admin provisioning, dispute arbitration, and financial refund execution.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-ADM-01: Review Verifications	Platform Admin	Firebase & Supabase	Approves or rejects pending shop registrations and courier applications. Approval instantly updates Firebase custom claims.
UC-ADM-02: Manage Admin Team	Platform Admin	Security Service	Provisions new administrators via cryptographic `ADMIN_SIGNUP_KEY` or consensus approval.
UC-ADM-03: Suspend / Reactivate Partner	Platform Admin	Firebase Auth	Disables or re-enables rogue shop owners or couriers, disabling their Firebase account and revoking platform tokens.
UC-ADM-04: Confirm Account Deletions	Platform Admin	Supabase DB	Reviews removal requests from couriers and shop owners; confirms permanent profile deactivation once all financial liabilities clear.
UC-ADM-05: Adjudicate Disputes & Refunds	Platform Admin	Razorpay Gateway	Investigates customer and courier dispute cases; authorizes electronic refund callbacks via Razorpay gateway adapter.

Subsystem 7: Immutable Security & Transaction Audit Trail

SECURITY AUDIT

Provides append-only, tamper-evident security logging for all critical platform lifecycle events, state transitions, and administrative overrides.

USE CASE ID & NAME	PRIMARY ACTOR	SECONDARY ACTOR	FUNCTIONAL DESCRIPTION & UML RELATIONSHIPS
UC-AUDIT-01: Record Audit Event	Audit Service	Supabase DB	Automatically writes structured audit record (action type, target entity, actor identity ID, payload JSON, and timestamp) on all critical mutations.
UC-AUDIT-02: Inspect Audit Trail Logs	Platform Admin	Audit Service	Filtered administrative inspection of historical logs by date range, action type (e.g. `ORDER_PLACED`, `SHOP_APPROVED`), or actor.

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Master Use Case Catalog & Traceability Matrix (Part 1)

Matrix: Functional Requirements Traceability
Coverage: UC-AUTH-01 through UC-ORD-05

8. Master Use Case Catalog & Traceability Matrix

Defines formal preconditions, triggers, primary flows, postconditions, and UML stereotyping for all platform capabilities:

UC ID	USE CASE NAME	ACTORS	PRECONDITIONS	TRIGGER & MAIN FLOW	UML RELATIONSHIPS
UC-AUTH-01	Sign Up with Role	Guest • Firebase	Email not registered.	User submits role, email, password; Firebase creates user; backend registers profile.	–
UC-AUTH-02	Sign In & Token Sync	User • Firebase	Registered account exists.	User provides credentials; Firebase issues ID token; backend resolves profile.	–
UC-AUTH-03	Enforce Verification Gate	Guard • Supabase	User authenticated.	System checks `verificationStatus`. If unapproved, locks route to `/verification`.	–
UC-AUTH-04	Request Account Removal	User • Admin	No active orders in transit.	User inputs offboarding reason; system registers deletion request awaiting admin review.	–
UC-MKT-01	Browse Approved Shops	Buyer / Guest	None (Public access).	System queries approved shops and displays storefronts with open/closed status.	–
UC-MKT-02	Search Food & Shops	Buyer / Guest	None (Public access).	User enters query; backend searches across item names, vendor names, and locations.	–
UC-MKT-03	View Personalized Feed	Buyer	Buyer logged in.	Recommendation service ranks products by user taste profile or executes cold-start.	–
UC-ORD-01	Manage Single-Shop Cart	Buyer	Items from single shop.	Buyer adds/removes items; client enforces single-shop invariant & ₹300 free delivery.	–
UC-ORD-02	Place Order (Checkout)	Buyer • Razorpay	Cart has ≥ 1 item.	Buyer confirms address; system reserves stock, validates payment, generates OTP.	`«include»` UC-ORD-02a `«include»` UC-ORD-02b
UC-ORD-02a	Reserve Stock Atomically	System • Supabase	Order initiated.	Pessimistic DB lock decrements product inventory. Fails if insufficient stock.	Included by UC-ORD-02
UC-ORD-02b	Process Online Payment	Buyer • Razorpay	Order initiated.	Razorpay modal collects payment and returns cryptographic signature verification.	Included by UC-ORD-02
UC-ORD-03	Track Live Order Status	Buyer	Order placed.	Client polls status every 6s; displays tracking progression and OTP when out for delivery.	–
UC-ORD-04	Cancel Order	Buyer • Dialog	Status is PLACED / ACCEPTED.	Buyer inputs cancellation reason; system restores inventory and marks order CANCELLED.	`«extend»` UC-ORD-03
UC-ORD-05	Rate & Review Order	Buyer	Order status DELIVERED.	Buyer submits 1-5 rating & comment; system records review (max 1 review per order).	–

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Master Use Case Catalog & Traceability Matrix (Part 2)

Matrix: Functional Requirements Traceability
Coverage: UC-SHOP-01 through UC-AUDIT-02

8. Master Use Case Catalog & Traceability Matrix (Continued)

UC ID	USE CASE NAME	ACTORS	PRECONDITIONS	TRIGGER & MAIN FLOW	UML RELATIONSHIPS
UC-SHOP-01	Register & Configure Shop	Shop Owner • Supabase	Verified Shop Owner role.	Owner enters shop details, Leaflet pin, and uploads storefront banner to Supabase.	–
UC-SHOP-02	Manage Menu Catalog	Shop Owner • Supabase	Shop registered.	Owner creates/updates dishes, prices, inventory stock counters, and product photos.	–
UC-SHOP-03	Process Kitchen Orders	Shop Owner	Order received.	Owner advances order states: `ACCEPTED` → `PREPARING` → `READY`.	–
UC-SHOP-04	Toggle Storefront Status	Shop Owner	Shop registered.	Owner toggles Open/Closed switch; updates public storefront availability immediately.	–
UC-DEL-01	Browse Available Queue	Delivery Partner	Verified Courier role.	Client polls available unassigned orders in `ACCEPTED/PREPARING/READY` state.	–
UC-DEL-02	Claim Delivery Assignment	Courier • Supabase	Assignment unassigned.	Courier accepts order; system executes atomic write lock query preventing race conditions.	«include» UC-DEL-02a
UC-DEL-02a	Acquire Pessimistic DB Lock	System • Supabase	Claim initiated.	Applies row-level DB lock; returns `409 Conflict` if claimed by competing courier.	Included by UC-DEL-02
UC-DEL-03	Confirm Kitchen Pickup	Delivery Partner	Assignment accepted.	Courier confirms receipt at kitchen; assignment becomes `PICKED_UP`, order `OUT_FOR_DELIVERY`.	–
UC-DEL-04	Execute Delivery Handover	Courier • Buyer	Out for delivery.	Courier arrives at drop pin; validates customer's 6-digit OTP to complete order.	«include» UC-DEL-04a
UC-DEL-04a	Verify Buyer 6-Digit OTP	Courier • Buyer	Handover initiated.	Courier enters buyer's OTP. Successful verification marks order `DELIVERED`.	Included by UC-DEL-04
UC-DEL-05	Report Delivery Issue	Courier • Admin	Delivery in transit.	Courier reports unreachable customer or address issue; marks delivery failed.	«extend» UC-DEL-04
UC-DEL-06	Review History & Earnings	Delivery Partner	Courier logged in.	Courier inspects fulfilled trips and cumulative delivery fee earnings (₹30/trip).	–
UC-ADM-01	Review Verifications	Platform Admin	Partner submitted KYC.	Admin inspects shop/courier credentials; approves or rejects, syncing Firebase claims.	–
UC-ADM-02	Manage Admin Team	Platform Admin	Valid Admin session.	Admin provisions team members with `ADMIN_SIGNUP_KEY` or consensus approval.	–
UC-ADM-03	Confirm Account Deletions	Platform Admin	Removal requested.	Admin verifies no outstanding financial liabilities and authorizes account removal.	–
UC-ADM-04	Adjudicate Disputes & Refunds	Admin • Razorpay	Dispute filed.	Admin investigates dispute notes and triggers automated electronic refund via Razorpay.	–
UC-AUDIT-01	Record Audit Event	System • Supabase	Action performed.	System appends immutable event row with action type, actor ID,	–

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Master PlantUML & Mermaid UML 2.5 Source Code

Format: Formal Diagram Source Code
Tooling: VS Code • GitHub • Mermaid Live

9. Production Mermaid.js Use Case Diagram Code

The snippet below contains the complete production-grade Mermaid.js definition for embedding in markdown documentation, repositories, or documentation portals:

```
flowchart TB
    %% QUICKBITE FORMAL UML 2.5 USE CASE ARCHITECTURE
    Guest["👤 Guest (Unregistered)"]
    User["👤 Registered User <<abstract>>"]
    Buyer["🛒 Buyer"]
    ShopOwner["🍳 Shop Owner"]
    Courier["🚴 Delivery Partner"]
    Admin["🛡️ Platform Admin"]

    Buyer --|> User
    ShopOwner --|> User
    Courier --|> User
    Admin --|> User

    subgraph QuickBite_Platform [System Boundary: QuickBite Platform]
        subgraph Sub_Auth [1. Identity & Access Subsystem]
            UC_SignUp(["UC-AUTH-01: Sign Up with Role Details"])
            UC_SignIn(["UC-AUTH-02: Sign In & Refresh Token"])
            UC_Gate(["UC-AUTH-03: Enforce Verification Gate"])
            UC_DelReq(["UC-AUTH-04: Request Account Removal"])
        end
        subgraph Sub_Mkt [2. Marketplace Discovery Subsystem]
            UC_Browse(["UC-MKT-01: Browse Approved Shops"])
            UC_Search(["UC-MKT-02: Search Food & Locations"])
            UC_Feed(["UC-MKT-03: View Personalized / Cold-Start Feed"])
        end
        subgraph Sub_Order [3. Cart & Ordering Subsystem]
            UC_Cart(["UC-ORD-01: Manage Single-Shop Cart"])
            UC_Checkout(["UC-ORD-02: Place Order (Checkout)"])
            UC_ResStock(["UC-ORD-02a: Reserve Stock Atomically"])
            UC_Pay(["UC-ORD-02b: Process Online Payment"])
            UC_Track(["UC-ORD-03: Track Live Order Status"])
            UC_Cancel(["UC-ORD-04: Cancel Order"])
            UC_OTP(["UC-ORD-04a: Verify 6-Digit Delivery OTP"])
            UC_Rate(["UC-ORD-05: Rate & Review Order"])
        end
        subgraph Sub_Shop [4. Kitchen & Shop Operations]
            UC_RegShop(["UC-SHOP-01: Register & Configure Shop"])
            UC_Menu(["UC-SHOP-02: Manage Menu Catalog"])
            UC_Kitchen(["UC-SHOP-03: Process Kitchen Orders"])
            UC_Toggle(["UC-SHOP-04: Toggle Storefront Status"])
        end
        subgraph Sub_Delivery [5. Courier Logistics & Fulfillment]
            UC_Avail(["UC-DEL-01: Browse Available Deliveries"])
            UC_Claim(["UC-DEL-02: Claim Delivery Assignment"])
            UC_Lock(["UC-DEL-02a: Acquire Pessimistic DB Lock"])
            UC_Pickup(["UC-DEL-03: Confirm Kitchen Pickup"])
            UC_Deliver(["UC-DEL-04: Execute Delivery Handover"])
            UC_Fail(["UC-DEL-05: Report Delivery Issue"])
            UC_Earnings(["UC-DEL-06: Review History & Earnings"])
        end
        subgraph Sub_Admin [6 & 7. Platform Governance & Audit]
            UC_Approve(["UC-ADM-01: Review Verifications & Approvals"])
            UC_AdminKey(["UC-ADM-02: Manage Admin Team"])
            UC_ConfDel(["UC-ADM-03: Confirm Account Deletions"])
            UC_Dispute(["UC-ADM-04: Adjudicate Disputes & Refunds"])
            UC_Audit(["UC-AUDIT-02: Inspect Audit Trail Logs"])
        end
    end

    subgraph External_Systems [Supporting External Actors]
        Ext_Firebase["🔥 Firebase Auth"]
        Ext_Razorpay["💳 Razorpay Gateway"]
        Ext_Supabase["🐘 Supabase Cloud"]
        Ext_Maps["🗺️ Google Maps"]
    end

    %% Actor Associations
    Guest --- UC_Browse & UC_Search & UC_SignUp
    User --- UC_SignIn & UC_DelReq
    Buyer --- UC_Browse & UC_Search & UC_Feed & UC_Cart & UC_Checkout & UC_Track & UC_Rate
    ShopOwner --- UC_RegShop & UC_Menu & UC_Kitchen & UC_Toggle
    Courier --- UC_Avail & UC_Claim & UC_Pickup & UC_Deliver & UC_Earnings
    Admin --- UC_Approve & UC_AdminKey & UC_ConfDel & UC_Dispute & UC_Audit
```